CLIMATE INTELLISENSE X

THE COMPLETE FARMING ESSENTIAL

TEAM GENESIS - DECODERS OF THE FARMING NIGHTMARES

Meet the Team



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CLIMATESENSE

AI & IoT-Based Smart Climate Intelligence for Farmers

Good morning everyone. We are the team "Ignited Mind" from PM SHRI Adarsha Vidyalaya, Mysore.

We have developed this idea to help farmers make better decisions about water, crops, and farming in a changing climate.

The Core Challenge Facing Modern Farmers

Climate Change Pressures

- Irregular rainfall patterns that disrupt farming
- Droughts leading to severe dry conditions
- Rising temperatures affecting crop viability
- Declining groundwater levels and acute water shortages

Risks of Farming by Guesswork

- No access to accurate, localized information
- Over-irrigation that wastes precious natural water resources
- Choosing unsuitable crops for the actual weather conditions
- High risk of total crop loss due to unexpected weather events

Action-Oriented Decision Support vs. Weather Info

Traditional Weather Info

Tells the farmer: "What the weather is"

Example Forecast:

“Rain may occur tomorrow.”

Provides raw meteorological indicators without context or practical action, leaving decisions completely to guesswork.

ClimateSense Decision Support

Tells the farmer: "What should I do?"

Actionable Recommendation:

“Postpone irrigation because rainfall is expected.”

Fuses IoT Sensors, AI, and Cloud Technology to continuously analyze temperature, humidity, rainfall, soil moisture, and groundwater level into simple, specific direct guidance.

challanges we faced and how we overcome them

Challenges Faced

- How can we collect accurate real-time information from the farm and convert it into simple advice for farmers?
- Making the system highly affordable and easy to use.
- Ensuring the system is suitable for rural areas and useful even for farmers who are completely unfamiliar with technology.

How We Overcome Them

- Low-Cost IoT Hardware: Built with low-cost sensors connected to a standard, accessible ESP32/Arduino microcontroller.
- Cloud Connectivity: Sensors continuously collect field data and seamlessly transfer it up to the cloud platform.
- Advanced AI Engine: Cloud AI analyses the real-time data alongside meteorological forecasts and historical climate patterns.
- Simplified Action: Complicated data is translated into simple, plain recommendations for the farmers.

TheSolution

Step 1 – Sensors
Sensors collect temperature, humidity, rainfall, soil moisture, and groundwater data.

Step 2 – ESP32
The ESP32 microcontroller collects the sensor information in the field.

Step 3 – Cloud
The collected field data is securely transmitted and sent to the cloud.

Step 4 – AI Analysis
AI analyses the farm data along with weather information and historical climate patterns.

Step 5 – Recommendations
The system generates drought alerts, irrigation advice, and crop suggestions.

Step 6 – Farmer
The farmer receives simple, direct actions through a mobile app or dashboard.

The Triple-Impact of ClimateSense

Environmental Impact

- Saves critical water resources
- Protects underground groundwater supplies
- Promotes sustainable, balanced farming techniques

Economic Impact

- Reduces crop losses from weather shocks
- Cuts unnecessary irrigation pumping costs
- Improves overall farm productivity and farmer income

Social Impact

- Helps farmers understand climate risks easily
- Supports rural farming communities
- Contributes to regional food security and stability

Future Roadmap: Smarter Farming Ecosystem

Future Capabilities

- Satellite data integration for macro insights
- Pest and disease prediction modeling
- AI-based crop yield prediction systems
- Multilingual voice assistance in regional languages
- SMS alerts for farmers without smartphones
- Automatic solenoid valve irrigation control
- Carbon footprint tracking and monitoring

“With the help of Artificial Intelligence and IoT, technology can become a powerful partner for agriculture.”

Our Message:

“ Sense the Climate.
Save Water.
Grow Smarter.”

THANK YOU!

FOR THE TIME YOU GAVE